Factors and levels of the fatigue test programme (MIM Ti-6Al-7Nb)

Factor	Level 1	Level 2	Level 3
Powder	TiH_2-derived	HDH	gas-atomised
Specimen geometry	round bar	flat	notched
Space-holder content	0 vol%	15 vol%	30 vol%
Oxygen content	0.15 wt%	0.30 wt%	0.45 wt%
Load ratio R	−1	0.1	0.5

Level combinations that cannot be produced or tested

Level of one factor	Level of another factor	Reason
Powder = TiH_2-derived	Space-holder content = 15 vol%	foam collapses on dehydrogenation
Powder = TiH_2-derived	Space-holder content = 30 vol%	foam collapses on dehydrogenation
Powder = HDH	Oxygen content = 0.15 wt%	HDH powder too high in oxygen
Powder = gas-atomised	Specimen geometry = flat	feedstock does not fill the mould
Specimen geometry = round bar	Oxygen content = 0.30 wt%	no stock